About Us

HESCO for Water Integrated Systems is an Egyptian engineering company specializing in the design, manufacturing, supply, installation, operation, and maintenance of water treatment systems and cooling towers. Established in 2026, the company was built upon the extensive experience of a highly qualified engineering and technical team that has successfully delivered projects with leading companies across the water treatment and industrial sectors.

Within a short period, **HESCO** has successfully become a partner in the success of several prestigious organizations, including Egyptian Steel, Misr Beni Suef Cement, Alumil Egypt, Domty, EIPICO Pharmaceuticals, UNI-CHARM Egypt, and The St. Regis New Capital, reflecting our commitment to delivering reliable, high-quality engineering solutions and building long-term customer relationships.

At **HESCO**, we believe that technical excellence, innovation, and customer satisfaction are the foundation of every successful project. Our mission is to provide reliable, energy-efficient, and cost-effective solutions that meet the highest engineering standards while addressing the unique requirements of each client.

Our expertise covers a comprehensive range of water treatment technologies, including Reverse Osmosis (RO), Ultrafiltration (UF), Media Filtration Systems, Water Softeners, Deionization (DI), Electrodeionization (EDI), Drinking Water Treatment Plants, Industrial Wastewater Treatment Plants, and Sewage Treatment Plants.

In addition to water treatment solutions, **HESCO** specializes in the design and manufacturing of high-performance cooling towers. Our product range includes Open Circuit Crossflow and Counterflow Cooling Towers manufactured in Fiberglass (FRP), Hot-Dip Galvanized Steel, and Stainless Steel, as well as Closed Circuit Cooling Towers designed for industrial and commercial applications. We also manufacture and supply all major cooling tower components, including fills, drift eliminators, louvers, nozzles, fans, and mechanical accessories.

We are committed to delivering sustainable engineering solutions, dependable after-sales support, and exceptional value that enable our clients to improve operational efficiency, reduce operating costs, and achieve long-term reliability.

Our Core Values

- Quality
- Integrity
- Innovation
- Customer Satisfaction
- Sustainability
- Continuous Improvement
- Safety

Our Mission

To deliver high-quality engineering solutions by combining advanced technologies, professional expertise, and outstanding customer support, while maintaining the highest standards of safety, quality, and environmental responsibility.

Our vision

to become one of the leading providers of integrated water and cooling solutions in Egypt while expanding our presence across the Middle East and Africa through continuous innovation, uncompromising quality, and long-term customer partnerships.

Services

1. **Reverse Osmosis (RO) Systems**

Introduction

Reverse Osmosis (RO) is one of the most advanced water purification technologies. It removes dissolved salts, minerals, bacteria, viruses, and contaminants through semi-permeable membranes, producing high-quality water suitable for industrial and drinking applications.

Applications

- Drinking water plants
- Boiler feed water
- Pharmaceutical industry
- Food & Beverage
- Hotels & Resorts

- Hospitals
- Manufacturing plants

2. **Ultrafiltration (UF) Systems**

Introduction

Ultrafiltration is a membrane filtration technology used to remove suspended solids, bacteria, viruses, and colloidal particles while maintaining high water flow with low energy consumption.

Applications

- Pretreatment for RO
- Drinking water
- Wastewater reuse
- Surface water treatment
- Industrial process water

3. **Media Filtration Systems**

Introduction

Media filtration systems utilize specialized filter media such as silica sand, activated carbon, anthracite, manganese media, and multimedia layers to remove suspended solids, turbidity, color, odor, chlorine, iron, and manganese.

Applications

- Pretreatment for RO
- Cooling towers
- Industrial water
- Irrigation
- Hotels
- Municipal water

4. **Water Softener Systems**

Introduction

Water softeners remove calcium and magnesium hardness through ion exchange resin, preventing scale formation and extending equipment life.

Applications

- Boilers
- Cooling towers
- Hotels

- Hospitals
- Laundry facilities
- Industrial production

5. **Deionization (DI) Systems**

Introduction

DI systems remove dissolved ions using cation and anion exchange resins to produce high-purity water.

Applications

- Laboratories
- Pharmaceutical plants
- Electronics manufacturing
- Chemical industries

6. **Electrodeionization (EDI) Systems**

Introduction

EDI combines ion exchange resins with electrical current to continuously produce ultrapure water without chemical regeneration.

Applications

- Pharmaceutical industry
- Semiconductor manufacturing
- Power plants
- High-purity process water

7. **Sewage Treatment Plants (STP)**

Introduction

Sewage Treatment Plants treat domestic wastewater using physical, biological, and chemical processes, allowing safe discharge or reuse.

Applications

- Residential compounds
- Hotels
- Hospitals
- Universities
- Commercial buildings
- Resorts

8. Industrial Wastewater Treatment Plants (WWTP)

Introduction

Industrial wastewater treatment systems are engineered to remove contaminants generated by industrial processes and comply with environmental regulations.

Applications

- Food industries
- Pharmaceutical plants
- Chemical factories
- Textile industry
- Paper mills
- Steel industry
- Cooling Towers

9. **Open Circuit Crossflow Cooling Towers**

Introduction

Crossflow cooling towers distribute hot water over fill media while air flows horizontally through the tower, providing efficient cooling with low pumping head and simplified maintenance.

Applications

- HVAC systems
- Hotels
- Commercial buildings
- Food factories
- Industrial plants

10. **Open Circuit Counterflow Cooling Towers**

Introduction

Counterflow cooling towers move air upward against the downward flow of water, achieving high thermal efficiency with a compact footprint.

Applications

- Power plants
- Steel industry
- Chemical plants
- Plastic industry
- Heavy manufacturing

11. **Closed Circuit Cooling Towers**

Introduction

Closed Circuit Cooling Towers circulate process fluid inside coils while cooling water and air remove heat externally. This keeps the process fluid clean and isolated from contamination.

Applications

- Ammonia refrigeration systems
- Chillers
- Food processing
- Pharmaceutical plants
- Plastic injection
- Chemical industry
- Industrial cooling processes

Why Choose HESCO?

- Highly experienced engineering and technical team.
- Complete design, manufacturing, installation, operation, and maintenance services.
- High-quality materials and internationally recognized components.
- Customized solutions tailored to customer requirements.
- Competitive pricing without compromising quality.
- Strong after-sales service and technical support.
- Commitment to sustainability, efficiency, and long-term customer satisfaction.
- Vision to become a leading engineering company in Egypt, the Middle East, and Africa.

Reverse Osmosis Systems

Reverse Osmosis is an advanced water purification technology that uses semi-permeable membranes to remove dissolved salts, minerals, organic matter, and impurities from water, producing high-quality purified water suitable for industrial and commercial applications.

At **HESCO** Water Integrated Systems, we provide complete solutions using a wide range of international and local brands. Our selection of equipment and components is based on each client's technical requirements, project conditions, budget considerations, and operational objectives to ensure reliable and efficient performance.

Main components

Raw Water Tank

Raw Water Pump

Dosing Systems

Multimedia Filter

Cartridge Filtration

High Pressure Pump

Instrumentations to control

RO Membranes

Control panel

Product Water Tank

Component	Function	Part of Brands
Membrane	Salt rejection & purification	Filmtec / Dupont / Toray / Hydranautics / LG
High Pressure Pump / Feed Pump	Provides required pressure	Grundfos / Lowara / CNP / Ebara
Pressure Vessels	Holds membranes	Codeline / Wave Cyber / phoenix / frotec
Instruments	Monitoring	+GF+ / Endress+Hauser / ABB / Siemens

Electrical Panel	Control system	Schneider / Siemens / ABB
Dosing Systems	Inject and dose chemicals into system	Etatron / EMEC
Cartridges	Remove Fine Suspended Particles	Pentair / Hydranautics / frotec / Local

Our Scope

HESCO provides:

- Engineering & Design
- Manufacturing
- Installation
- Commissioning
- Operation & Maintenance
- Chemical Cleaning
- Spare Parts Supply

Wastewater Treatment Plants

Wastewater Treatment is an integrated process designed to remove contaminants, organic matter, suspended solids, and harmful substances from industrial and domestic wastewater to produce treated water suitable for safe discharge, reuse, or recycling.

HESCO provides customized wastewater treatment solutions for different industrial sectors including food & beverage, pharmaceutical, chemical, textile, and manufacturing industries.

At HESCO Water Integrated Systems, we provide complete wastewater treatment solutions using a wide range of international and local brands. Our systems are engineered according to each client's wastewater characteristics, required treatment standards, operational requirements, and project objectives to achieve efficient, reliable, and sustainable performance.

Typical wastewater treatment process:

1- Screening

Removal of large suspended materials and solid particles.

2- Equalization Tank

Balancing flow and wastewater characteristics before treatment.

3- Biological Treatment

Using microorganisms to reduce organic load (COD & BOD).

4- Clarification

Separation of biological sludge from treated water.

5- Filtration

Removal of remaining suspended particles.

6- Disinfection

Elimination of microorganisms before reuse or discharge.

Main Components & part of Brands

Component	Function	Part of Brands
Pumps	Wastewater transfer	Grundfos / Lowara / Ebara / CNP
Blowers	Oxygen supply for biological treatment	AERZEN / Robuschi / FPZ
Diffusers	Air distribution	EDI / SSI / Daysil
Mechanical Screens	Solid separation	Huber / Local Manufacturing
Control Panels	Automation	Siemens / Schneider / ABB
Instruments	Monitoring	+GF+ / Endress+Hauser / ABB

Cooling Towers

A Cooling Tower is a heat rejection system designed to remove heat from industrial process water by transferring heat to atmospheric air through evaporation and direct contact between water and air.

Cooling towers are widely used in industrial plants, HVAC systems, power generation, and manufacturing processes.

At **HESCO** Water Integrated Systems, we design, manufacture, and supply cooling tower solutions using different international and local technologies. We select suitable designs, materials, and components according to each project's thermal requirements, operating conditions, and client expectations.

Cooling tower operation depends on:

- Hot water enters the tower distribution system.
- Water flows through the fill material.
- Fans move air through the tower.
- Heat transfers from water to air through evaporation.
- Cooled water returns back to the process.
- Cooling Tower Types
- Open Circuit Cooling Towers
- Closed Circuit Cooling Towers
- Counter Flow Cooling Towers
- Cross Flow Cooling Towers
- FRP Cooling Towers
- Galvanized Steel Cooling Towers

Main Components & part of Brands

Component	Function	Part of Brands
Fan	Air movement	Wing fan / Local
Motor	Fan driving	Siemens / ABB / Gamak
Fill Media	Heat transfer	Local (Our Factory)
Drift Eliminator	Reduce water loss	Local (Our Factory)
Spray Nozzles	Water distribution	Marley / Local
Structure	Tower body	FRP / Galvanized Steel / Stainless steel

Filtration Systems

Filtration systems are designed to remove suspended solids, turbidity, organic materials, chlorine, and other impurities from water to improve quality and protect downstream equipment.

Water passes through different filtration stages depending on application:

Multimedia Filtration

Sand Filtration

Activated Carbon Filtration

Cartridge Filtration

Bag Filtration

At **HESCO** Water Integrated Systems, we provide complete filtration solutions using selected international and local brands based on water quality analysis, required filtration efficiency, and customer application.

Main Components & Part of Brands

Component	Function	Part of Brands
Filter Vessels	Contain Filtration Media (Quartz Sand / Activated Carbon)	Pentair / Canature / Greenway / Local
Valves	Control, regulate, Isolate, and direct water flow within the treatment system	Fleck / Runxin / Clack / Automatic butterfly valves
Cartridges	Remove Fine Suspended Particles	Pentair / Hydranautics / frotec / Local
Pumps	water transfer	Grundfos / Lowara / Ebara / CNP

DI & EDI Systems

Demineralization & EDI Systems

DI (Deionization) and EDI (Electrodeionization) systems are advanced purification technologies used to remove dissolved ions and produce high-purity water.

DI:

Water passes through cation and anion exchange resins.

Dissolved ions are removed.

EDI:

Uses electrical current and membranes to continuously remove ions without chemical regeneration.

At **HESCO** Water Integrated Systems, we provide high-purity water solutions using advanced DI and EDI technologies with international and local components selected according to required water quality standards.

Main Components & Part of Brands

Component	Function	Part of Brands
EDI Modules	To continuously remove dissolved ions from pre-treated water using an electrically driven ion exchange process	Suez / Veolia / SnowPure
Ion Exchange Resin	Remove dissolved ions and impurities from water through an ion exchange process	Purolite / Lanxess
Membrane	Salt rejection & purification	Filmtec / Dupont / Toray / Hydranautics / LG
Pumps	water transfer	Grundfos / Lowara / Ebara / CNP
Instruments	Monitoring	+GF+ / Endress+Hauser / ABB / Siemens

Chemical Cleaning & Descaling

Chemical Cleaning Services

Chemical cleaning is a controlled process used to remove accumulated scale, deposits, biological growth, corrosion products, oils, and other contaminants from water treatment and industrial equipment.

It is commonly applied to:

- RO Membranes
- Heat Exchangers
- Cooling Towers
- Chillers
- Boilers
- Filtration Systems
- Industrial Piping

The cleaning process generally includes:

1. System Inspection
Assessment of the equipment condition and identification of the type of fouling.
2. Chemical Selection
Selection of the appropriate cleaning chemical based on the contaminant and equipment material.
3. Preparation & Isolation
The equipment is isolated and connected to the chemical cleaning system.
4. Chemical Circulation
The cleaning solution is circulated through the equipment under controlled conditions.
5. Flushing
The system is thoroughly flushed with clean water to remove chemical residues.
6. Final Inspection & Testing
System performance is checked before returning the equipment to normal operation

At **HESCO** Water Integrated Systems, we provide professional chemical cleaning solutions using a wide range of international and local chemicals and equipment. Our cleaning programs are selected according to the type of contamination, equipment material, water quality, and operating conditions to achieve effective cleaning while protecting system components.

Our Chemical Cleaning Solutions

- RO Membrane Cleaning
- Cooling Tower Descaling
- Heat Exchanger Cleaning
- Chiller Cleaning
- Boiler Cleaning
- Pipeline Cleaning
- Filter Media Cleaning

Operation & Maintenance

Operation & Maintenance Services

Operation & Maintenance (O&M) is the continuous management of water treatment and cooling systems to ensure safe operation, stable performance, high efficiency, and extended equipment lifetime.

Our O&M approach combines:

Operation + Preventive Maintenance + Corrective Maintenance + Performance Monitoring

Our O&M program typically includes:

1. **Daily Operation Monitoring**

- Flow rates
- Pressure
- Temperature
- pH
- Conductivity
- Water quality

2. **Preventive Maintenance**

Scheduled inspection and servicing of:

- Pumps
- Motors
- Valves
- Filters
- Membranes
- Electrical panels
- Instrumentation

3. **Corrective Maintenance**

Rapid troubleshooting and repair of equipment failures and abnormal operating conditions.

4. **Performance Monitoring**

Monitoring system KPIs to identify performance degradation and optimize operation.

5. **Reporting**

Providing technical reports including:

- Operating parameters
- Maintenance activities
- Chemical consumption
- Spare parts
- Equipment condition
- Recommendations

Our O&M Services

- Water Treatment Plant Operation
- RO Plant Operation
- Wastewater Plant Operation
- Cooling Tower Operation
- Preventive Maintenance
- Corrective Maintenance
- Chemical Management
- Spare Parts Management
- Performance Optimization
- Technical Reporting
- Emergency Technical Support

Spare Parts & Technical Support

At **HESCO** Water Integrated Systems, we supply spare parts from a wide range of international and local brands, allowing our clients to select the most suitable solution according to equipment specifications, availability, budget, and operational requirements.

At **HESCO** Water Integrated Systems, we provide comprehensive operation and maintenance services using qualified technical teams, reliable spare parts, and internationally recognized equipment and component brands. Our maintenance programs are customized according to each facility's operating conditions, equipment requirements, and performance targets.

Reliable spare parts availability is essential for maintaining the continuity and efficiency of water treatment and industrial systems.

HESCO provides original, equivalent, and locally manufactured components depending on the equipment requirements and client preferences.

Depending on the equipment and application, we can work with brands such as:

Grundfos – Lowara – Ebara – Siemens – ABB – Schneider – Danfoss – GEMU – Bray – Endress+Hauser – WIKA – Filmtec – DuPont – Toray – Hydranautics – LG – Fleck – Clack – Pentair – SEKO – ProMinent

Water Analysis & Consultation

Water Analysis & Technical Consultation

Water analysis is the first step in designing an effective treatment system.

Laboratory and field analysis can identify parameters such as:

- pH
- TDS
- Conductivity
- Hardness
- Alkalinity
- Turbidity
- Chlorine
- Iron
- Manganese
- Silica
- COD
- BOD
- TSS
- Microbiological Parameters

At HESCO Water Integrated Systems, we provide technical consultation based on detailed water analysis, process requirements, and site conditions. We work with international and local technologies and brands to develop the most technically and economically suitable solution for each client.

Our technical approach includes:

1. Water Sampling

Collection of representative water samples.

2. Laboratory Analysis

Testing the required physical, chemical, and biological parameters.

3. Technical Assessment

Evaluation of the water quality against the intended application.

4. System Selection

Selection of the appropriate treatment technology.

5. Process Design

Development of the treatment process and equipment requirements.

6. Technical & Commercial Proposal

Providing a complete solution including equipment, scope of supply, performance requirements, and commercial considerations.

Our Consultation Covers

- New Water Treatment Plants
- RO Systems
- Wastewater Treatment
- Cooling Tower Systems
- Filtration Systems
- Softener Systems
- DI & EDI
- Water Reuse & Recycling
- Plant Upgrades
- Capacity Expansion
- Troubleshooting & Optimization